

The Stochastic Discount Factor and the CAPM: Arbitrage, Beta Pricing and the Consumption Puzzles

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Abstract

This note develops asset pricing from a single equation, $p = \mathbb{E}[mx]$, and shows that the Capital Asset Pricing Model is one special case of it. Section 2 proves that absence of arbitrage is equivalent to the existence of a strictly positive stochastic discount factor. Section 3 derives the risk premium, the beta representation, the Hansen–Jagannathan bound and the mean–variance frontier from that equation alone. Section 4 shows that any linear factor model is equivalent to a beta representation, and that the CAPM is the case in which the single factor is the return on aggregate wealth; the consumption CAPM is the case in which the factor is consumption growth. Section 5 calibrates the consumption model to standard post-war moments and reports the two failures this produces: matching the equity premium requires a coefficient of relative risk aversion of 25.8 at the Hansen–Jagannathan bound and 133.1 once the observed consumption–return correlation is imposed, and either value implies a risk-free rate of tens of per cent against an observed one per cent. Section 6 implements the theory numerically, verifying the pricing equation by Monte Carlo to within one standard error and demonstrating on a simulated cross-section that a single-factor model leaves economically large pricing errors when a second priced factor is omitted (GRS statistic 3.61, $p = 1.1 \times 10^{-4}$), which the correct two-factor model removes. All computations are in Python and reproduced in Appendix A.

Notation

Table 1: Notation fixed for the whole note.

Symbol	Meaning (and where introduced)
$(\Omega, \mathcal{F}, \mathbb{P})$	probability space; $\mathbb{P}$ the physical measure
S	number of states of the world in the finite-state market (Section 2)
π_s	physical probability of state s
$x, \underline{X}$	payoff, and the space of traded payoffs (Definition 2.1)
$p(x)$	price today of the payoff x (Definition 2.2)
q_s, Q_s	state price of state s (Theorem 2.4)
m	stochastic discount factor, or pricing kernel (Definition 2.5)
$R^i = x^i/p^i$	gross return on asset i ; R^e a gross excess return
$R^f = 1/\mathbb{E}[m]$	gross risk-free return; r^f its log
R^W	gross return on aggregate wealth (the market return)
$\beta_{i,f}$	regression coefficient of R^i on the factor f (Proposition 3.2)
λ_f	price of risk attached to the factor f
SR	Sharpe ratio $\mathbb{E}[R^e]/\sigma(R^e)$
$u(\cdot), \delta$	period utility, subjective discount factor (Section 4)
γ	coefficient of relative risk aversion
Δc	log consumption growth $\ln(C_1/C_0)$; $g = \mathbb{E}[\Delta c]$, $\sigma_c = \sigma(\Delta c)$
σ_m, ρ_{cm}	volatility of the log market return; its correlation with Δc
α_i	pricing error of asset i under a stated factor model (Section 6)

1 Introduction

Asset pricing is usually taught as a sequence of models: mean–variance portfolio theory, then the CAPM, then multifactor extensions, then the consumption-based model. Presented that way the models look like competitors, and the natural question is which one is right. It is a misleading way to organise the subject. Every model in that list is the same equation,

$$p = \mathbb{E}[mx], \tag{1.1}$$

with a different specification of m . The disagreement between them is not about the structure of pricing but about what drives marginal utility.

This note takes (1.1) as the object of study. Three claims are established, and it is worth stating them before the machinery.

1. **Existence of m is exactly no arbitrage.** A strictly positive stochastic discount factor exists if and only if the market admits no arbitrage (Theorem 2.6). Nothing about preferences, equilibrium or rationality is required. Assuming a positive pricing kernel in empirical work is therefore no stronger than assuming that money is not lying on the pavement.
2. **Beta pricing is a rewriting, not a theory.** Any linear factor model $m = a + b'f$ is equivalent to an expected-return–beta representation, and conversely (Theorem 4.2). The CAPM’s content is not the linear relation between expected return and beta, which is close to vacuous; it is the specific claim that the return on aggregate wealth is the factor.
3. **The failure is quantitative, and it is severe.** When m is given economic content as the marginal rate of substitution of a representative consumer, the model does not miss by a little. Reconciling it with post-war US moments requires risk aversion two orders of magnitude above the values that survive introspection or experiment, and the same parameter then predicts a risk-free rate of tens of per cent (Section 5).

Roadmap. Section 2 constructs the pricing operator in a finite-state market and proves the fundamental theorem. Section 3 extracts from (1.1) the risk premium, the beta representation, the Hansen–Jagannathan bound and the mean–variance frontier. Section 4 specialises m : first to a linear factor model in general, then to the CAPM and the consumption CAPM, with the conditions each requires stated explicitly. Section 5 calibrates and reports the equity premium and risk-free rate puzzles. Section 6 implements everything numerically. Section 7 discusses the responses in the literature, and Section 8 concludes. Appendix A contains the Python code that produced every number and figure below.

References used in this note. The works below are those this note draws on, with the use made of each.

- [Co] Cochrane, *Asset Pricing* (2005). The principal source. The organisation around $p = \mathbb{E}[mx]$, the equivalence of linear factor models and beta pricing in Section 3 and Section 4, and the discussion of the consumption puzzles in Section 5 follow his treatment.
- [Du] Duffie, *Dynamic Asset Pricing Theory* (2001). Used for the finite-state fundamental theorem of Section 2, stated there in the generality needed here.
- [Ro] Ross, A simple approach to the valuation of risky streams (1978). The original separating-hyperplane argument behind Theorem 2.6.
- [HK] Harrison and Kreps (1979). The martingale-measure formulation of the same equivalence, cited in Remark 2.8.

- [HJ] Hansen and Jagannathan, Implications of security market data for models of dynamic economies (1991). The volatility bound of Theorem 3.3 and the diagnostic use made of it in Section 5.
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- [Sh] Sharpe (1964) and [Li] Lintner (1965). The original CAPM, recovered here as Corollary 4.5.
- [Br] Breeden (1979). The consumption CAPM of Section 4.3.
- [GRS] Gibbons, Ross and Shanken (1989). The joint test of zero pricing errors used in Section 6.
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- [CC] Campbell and Cochrane (1999), [BY] Bansal and Yaron (2004), [EZ] Epstein and Zin (1989). The three standard responses surveyed in Section 7.
- [Ca] Campbell, Consumption-based asset pricing (2003). The source of the moment magnitudes used in the calibration of Table 2.

2 No Arbitrage and the Existence of a Discount Factor

2.1 The market

We work in a two-date, finite-state market. Today is date 0; tomorrow is date 1. Uncertainty is described by a finite set of states $s \in \{1, \dots, S\}$, exactly one of which is realised, with strictly positive physical probabilities π_s .

Definition 2.1 (Payoff space). A payoff is a vector $x = (x_1, \dots, x_S) \in \mathbb{R}^S$, where x_s is the number of units of account delivered in state s . The set of payoffs available by trading is the *payoff space* $\underline{X} \subseteq \mathbb{R}^S$. Because investors may form portfolios freely, $\underline{X}$ is a linear subspace: if $x, z \in \underline{X}$ and $a, b \in \mathbb{R}$ then $ax + bz \in \underline{X}$.

Definition 2.2 (Price functional). The price functional is the map $p : \underline{X} \rightarrow \mathbb{R}$ assigning to each traded payoff its date-0 price. It is assumed linear,

$$p(ax + bz) = ap(x) + bp(z), \quad x, z \in \underline{X}, \quad a, b \in \mathbb{R}. \quad (2.1)$$

Linearity is an assumption about market frictions, and it is worth naming what it rules out: bid–ask spreads, transaction costs, short-sale constraints, position limits and price impact. It delivers the law of one price, since two portfolios with the same payoff have a difference with zero payoff and hence zero price.

Definition 2.3 (Arbitrage). An *arbitrage* is a traded payoff $x \in \underline{X}$ with $x \geq 0$, $x \neq 0$ and $p(x) \leq 0$. The market is arbitrage-free if no such payoff exists, that is if

$$x \in \underline{X}, \quad x \geq 0, \quad x \neq 0 \implies p(x) > 0. \quad (2.2)$$

Linearity and (2.2) are the only two assumptions used in this section. No utility function, no equilibrium and no distributional assumption appears.

2.2 The fundamental theorem

Theorem 2.4 (Basic Arbitrage Theorem, complete-market case). *Suppose $\underline{X} = \mathbb{R}^S$, so that every state-contingent payoff is traded. Then the market is arbitrage-free if and only if there exists a strictly positive vector $q = (q_1, \dots, q_S) \in \mathbb{R}_{++}^S$ with*

$$p(x) = \sum_{s=1}^S q_s x_s \quad \text{for every } x \in \mathbb{R}^S. \quad (2.3)$$

The vector q is then unique, and $q_s = p(e_s)$ where e_s is the payoff paying one unit in state s and nothing otherwise.

Proof. ($\Leftarrow$) Suppose such a q exists. If $x \geq 0$ and $x \neq 0$ then $x_s > 0$ for at least one s , and since every $q_s > 0$ the sum in (2.3) is strictly positive. Hence $p(x) > 0$ and (2.2) holds.

($\Rightarrow$) Suppose the market is arbitrage-free. The vectors $e_1, \dots, e_S$ form a basis of $\mathbb{R}^S$, so any x can be written $x = \sum_s x_s e_s$, and linearity (2.1) gives

$$p(x) = \sum_{s=1}^S x_s p(e_s).$$

Setting $q_s := p(e_s)$ yields (2.3). Each e_s satisfies $e_s \geq 0$ and $e_s \neq 0$, so (2.2) gives $q_s = p(e_s) > 0$. Uniqueness is immediate: if q and $\tilde{q}$ both represent p then $(q - \tilde{q})' e_s = 0$ for every s , so $q = \tilde{q}$. $\square$

The quantities q_s are *state prices* or Arrow–Debreu prices: q_s is what one pays today for one unit of account delivered in state s and nothing elsewhere. Applying (2.3) to the riskless payoff $x_s = 1$ for all s gives

$$\frac{1}{R^f} = \sum_{s=1}^S q_s, \quad (2.4)$$

so state prices sum to the price of a zero-coupon bond.

Definition 2.5 (Stochastic discount factor). Given state prices q and physical probabilities π , the *stochastic discount factor* (SDF), or pricing kernel, is the random variable taking the value

$$m_s := \frac{q_s}{\pi_s} \quad (2.5)$$

in state s .

Substituting (2.5) into (2.3) gives the central pricing equation

$$p(x) = \sum_{s=1}^S \pi_s m_s x_s = \mathbb{E}[mx]. \quad (2.6)$$

The separation performed in (2.5) is the whole point: a state price mixes the probability of a state with the value of a unit of account in it, and dividing by π_s strips out the probability, leaving the valuation.

Theorem 2.6 (Fundamental theorem of asset pricing, finite state). *Let $\underline{X} \subseteq \mathbb{R}^S$ be a linear subspace and $p : \underline{X} \rightarrow \mathbb{R}$ linear. The market $(\underline{X}, p)$ is arbitrage-free if and only if there exists a strictly positive random variable $m > 0$ with $p(x) = \mathbb{E}[mx]$ for every $x \in \underline{X}$. If $\underline{X} = \mathbb{R}^S$ the SDF is unique; if $\underline{X} \subsetneq \mathbb{R}^S$ there are infinitely many such m , all agreeing on $\underline{X}$.*

Reference and remarks. Sufficiency is the argument already given: $m > 0$ and $x \geq 0$, $x \neq 0$ force $\mathbb{E}[mx] > 0$. Necessity in the complete case is Theorem 2.4. In the incomplete case the price functional is defined only on the subspace $\underline{X}$ and must be extended to $\mathbb{R}^S$ while preserving strict positivity; this is a separating-hyperplane argument, given by Ross [Ro] and set out in this form in Duffie [Du, Ch. 1]. For non-uniqueness, note that if m represents p on $\underline{X}$ and ε is any random variable orthogonal to $\underline{X}$, that is $\mathbb{E}[\varepsilon x] = 0$ for all $x \in \underline{X}$, then $m + \varepsilon$ represents p equally well, and for ε small enough it remains strictly positive. $\square$

Remark 2.7 (Existence, uniqueness, and what each corresponds to). Theorem 2.6 separates two properties that are often conflated. *Existence* of a positive SDF is equivalent to no arbitrage. *Uniqueness* is equivalent to market completeness. In an incomplete market prices are still consistent, but a claim outside $\underline{X}$ has a whole interval of arbitrage-free values, one for each admissible m . This is the discrete-time counterpart of the incompleteness that arises in continuous time whenever a second, untraded risk factor is present.

Remark 2.8 (The risk-neutral measure). Normalising the SDF by its mean produces a third representation. Define

$$\pi_s^* := \frac{q_s}{\sum_j q_j} = \pi_s \frac{m_s}{\mathbb{E}[m]}, \quad (2.7)$$

which is strictly positive and sums to one, hence is a probability measure. In terms of it, (2.6) reads $p(x) = \mathbb{E}^*[x]/R^f$: the price is the discounted expectation under the risk-neutral measure. State prices, the SDF and the risk-neutral measure are therefore three coordinate systems on one object, related by $q_s = \pi_s m_s = \pi_s^*/R^f$. The martingale formulation is due to Harrison and Kreps [HK]; nothing below depends on it, and we work with m throughout because it is the representation in which m can be given economic content.

3 The Central Pricing Equation and its Consequences

Everything in this section is a consequence of (2.6) and elementary properties of covariance. No further assumption is introduced.

3.1 Returns, the risk-free rate and the risk premium

For an asset with price $p^i > 0$ and payoff x^i , the gross return is $R^i = x^i/p^i$, and dividing (2.6) by p^i gives the equation in return form,

$$1 = \mathbb{E}[mR^i]. \quad (3.1)$$

Applying (3.1) to a riskless asset, whose return is known,

$$R^f = \frac{1}{\mathbb{E}[m]}. \quad (3.2)$$

Proposition 3.1 (Risk premium). *For any gross return R^i ,*

$$\mathbb{E}[R^i] - R^f = -R^f \operatorname{Cov}(m, R^i) = -\frac{\operatorname{Cov}(m, R^i)}{\mathbb{E}[m]}. \quad (3.3)$$

For any gross excess return $R^e = R^i - R^f$,

$$\mathbb{E}[mR^e] = 0 \quad \Longleftrightarrow \quad \mathbb{E}[R^e] = -R^f \operatorname{Cov}(m, R^e). \quad (3.4)$$

Proof. Expand (3.1) with the definition of covariance:

$$1 = \mathbb{E}[mR^i] = \mathbb{E}[m] \mathbb{E}[R^i] + \text{Cov}(m, R^i).$$

Divide by $\mathbb{E}[m] > 0$ and use (3.2) to get $R^f = \mathbb{E}[R^i] + R^f \text{Cov}(m, R^i)$, which rearranges to (3.3). For (3.4), apply (3.1) to R^i and R^j and subtract, giving $\mathbb{E}[mR^e] = 0$; expanding the covariance as before yields the stated form. $\square$

Equation (3.3) is the entire theory of risk premia in one line, and its content should be stated carefully. An asset earns a premium over the safe rate *only* to the extent that its return covaries negatively with the discount factor, that is only if it pays badly in states the investor regards as expensive. Variance enters nowhere except through this covariance. An asset with enormous variance that is uncorrelated with m earns exactly the risk-free rate. Idiosyncratic risk is not priced, and this holds without any of the CAPM's distributional or preference assumptions.

3.2 The beta representation

Proposition 3.2 (Beta representation). *Let m price all assets. Then for every gross return R^i ,*

$$\mathbb{E}[R^i] = R^f + \beta_{i,m} \lambda_m, \quad \beta_{i,m} := \frac{\text{Cov}(R^i, m)}{\text{Var}(m)}, \quad \lambda_m := -\frac{\text{Var}(m)}{\mathbb{E}[m]}. \quad (3.5)$$

The quantity $\beta_{i,m}$ is the slope coefficient in the regression of R^i on m , and λ_m is common to all assets and strictly negative.

Proof. Write (3.3) as

$$\mathbb{E}[R^i] - R^f = -\frac{\text{Cov}(m, R^i)}{\mathbb{E}[m]} = \underbrace{\frac{\text{Cov}(R^i, m)}{\text{Var}(m)}}_{\beta_{i,m}} \cdot \underbrace{\left(-\frac{\text{Var}(m)}{\mathbb{E}[m]}\right)}_{\lambda_m},$$

where the second equality multiplies and divides by $\text{Var}(m) > 0$. That $\beta_{i,m}$ is a regression coefficient is the definition of least squares; $\lambda_m < 0$ because $\text{Var}(m) > 0$ and $\mathbb{E}[m] = 1/R^f > 0$. $\square$

The structure of (3.5) is the one every empirical asset pricing model has: a quantity of risk, which is asset-specific, multiplied by a price of risk, which is not. The negative sign of λ_m is not a defect. Assets that covary positively with the discount factor are insurance, and insurance is expensive, so it earns less than the safe rate.

3.3 The Hansen–Jagannathan bound

The following is the single most useful diagnostic in the subject, because it converts a statement about asset prices into a testable restriction on any candidate discount factor without requiring the model to be specified.

Theorem 3.3 (Hansen–Jagannathan volatility bound). *Let R^e be any gross excess return, so that $\mathbb{E}[mR^e] = 0$. Then*

$$\frac{\sigma(m)}{\mathbb{E}[m]} \geq \frac{|\mathbb{E}[R^e]|}{\sigma(R^e)} = |\text{SR}|, \quad (3.6)$$

where SR denotes the Sharpe ratio of R^e . The bound holds with equality if and only if m and R^e are perfectly correlated.

Proof. From $\mathbb{E}[mR^e] = 0$ and the definition of covariance,

$$0 = \mathbb{E}[m]\mathbb{E}[R^e] + \text{Cov}(m, R^e) = \mathbb{E}[m]\mathbb{E}[R^e] + \text{Corr}(m, R^e) \sigma(m) \sigma(R^e).$$

Solving for the expected excess return,

$$\mathbb{E}[R^e] = -\text{Corr}(m, R^e) \frac{\sigma(m)}{\mathbb{E}[m]} \sigma(R^e).$$

Taking absolute values, dividing by $\sigma(R^e) > 0$ and using $|\text{Corr}(m, R^e)| \leq 1$ gives (3.6). Equality requires $|\text{Corr}(m, R^e)| = 1$, which is the stated condition. $\square$

Corollary 3.4 (The bound is a hurdle any model must clear). *A candidate discount factor is inconsistent with observed prices unless its coefficient of variation is at least as large as the largest Sharpe ratio available in the market. The bound involves no free parameters and no assumption about the model, so it can be applied before any estimation is attempted.*

Section 5 applies Corollary 3.4 to the consumption-based kernel and finds that it fails by an order of magnitude at plausible risk aversion. That single calculation is the equity premium puzzle.

3.4 The mean–variance frontier

The bound has a geometric counterpart that connects (2.6) to classical portfolio theory.

Proposition 3.5 (Frontier returns). *A return R^{mv} lies on the mean–variance frontier, in the sense that it has minimum variance among all returns with the same mean, if and only if it can be written*

$$R^{mv} = R^f + w \left(\frac{m - \mathbb{E}[m]}{\mathbb{E}[m]} \right)^\perp \quad (3.7)$$

for some scalar w , where the superscript denotes the projection of m onto the space of traded payoffs. Equivalently, frontier returns are exactly those perfectly correlated with the discount factor, and only for these does (3.6) hold with equality.

Reference. This is the classical duality between the mean–variance frontier and the discount factor; see Cochrane [Co, Ch. 5], where the projection $m^* = \text{proj}(m \mid \underline{X})$ is constructed explicitly and shown to price all traded assets. We use only the conclusion: the frontier is populated by returns maximally correlated with m , which is why the maximum Sharpe ratio in the market is attained on the frontier and why Theorem 3.3 binds there. $\square$

Remark 3.6 (Why mean–variance analysis survives). Proposition 3.5 explains an otherwise puzzling fact: mean–variance analysis is used constantly in practice although the quadratic utility that motivates it is indefensible. The reason is that the frontier is not really about quadratic utility at all. It is the geometry of the pricing equation, and it appears whenever (2.6) holds, whatever generates m .

4 From the Discount Factor to the CAPM

Sections 2 and 3 used no economics beyond no arbitrage. To obtain a model one must say what m is. This section does so in three steps of increasing specificity: linear factor models in general, the CAPM, and the consumption CAPM.

4.1 Linear factor models and beta pricing are the same thing

Definition 4.1 (Linear factor model). A *linear factor model* asserts that the discount factor is an affine function of a K -vector of factors $f = (f_1, \dots, f_K)'$,

$$m = a + b'f, \quad (4.1)$$

for constants $a \in \mathbb{R}$ and $b \in \mathbb{R}^K$.

Theorem 4.2 (Equivalence of factor models and beta representations). *Suppose $\text{Var}(f)$ is non-singular. The discount factor takes the form (4.1) and prices all assets if and only if expected returns satisfy the beta representation*

$$\mathbb{E}[R^i] = R^f + \beta_i' \lambda, \quad \beta_i := \text{Var}(f)^{-1} \text{Cov}(f, R^i), \quad (4.2)$$

with the price-of-risk vector

$$\lambda = -\frac{\text{Var}(f) b}{\mathbb{E}[m]} = -R^f \text{Var}(f) b. \quad (4.3)$$

Proof. ($\Rightarrow$) Suppose (4.1) holds. Since a is constant, $\text{Cov}(m, R^i) = b' \text{Cov}(f, R^i)$. Substituting into Proposition 3.1,

$$\mathbb{E}[R^i] - R^f = -\frac{b' \text{Cov}(f, R^i)}{\mathbb{E}[m]}.$$

Insert $\text{Cov}(f, R^i) = \text{Var}(f) \beta_i$, which is the definition of the multivariate regression coefficient in (4.2), to obtain

$$\mathbb{E}[R^i] - R^f = -\frac{b' \text{Var}(f) \beta_i}{\mathbb{E}[m]} = \beta_i' \lambda$$

with λ as in (4.3), using the symmetry of $\text{Var}(f)$.

($\Leftarrow$) Suppose (4.2) holds for some λ . Define $b := -\text{Var}(f)^{-1} \lambda / R^f$ and $a := 1/R^f - b' \mathbb{E}[f]$, and set $m = a + b'f$. Then $\mathbb{E}[m] = a + b' \mathbb{E}[f] = 1/R^f$, so this candidate prices the riskless asset correctly, and reversing the chain of equalities above shows that it reproduces (4.2) for every i , hence prices all assets. $\square$

Theorem 4.2 is worth pausing on, because it disposes of a common misunderstanding. Finding that expected returns line up with betas on some factor is *not* evidence for a theory. Any linear factor model produces such a relation, and any such relation can be rewritten as a linear factor model. Given enough freedom in choosing f , a beta representation can always be constructed; the extreme case takes the factor to be the return on the mean-variance efficient portfolio, which by Proposition 3.5 always exists. The economic content of a model lies entirely in the claim about *which* factor, specified in advance and independently of the returns to be explained.

Remark 4.3 (Sign of the price of risk). Equation (4.3) carries information often lost in empirical work: a model does not merely predict that λ is non-zero, it predicts its sign. A factor that raises marginal utility, so that $b > 0$ in (4.1), must carry a negative price of risk. Assets loading on it are hedges. Recovering a λ of the wrong sign is a rejection of the model, not a calibration detail.

4.2 The CAPM

Assumption 4.4 (CAPM environment). There are two dates. A representative investor consumes only at date 1, so that terminal consumption equals terminal wealth, $C_1 = W_1 = W_0 R^W$, where R^W is the gross return on aggregate wealth. Either (i) the investor has quadratic utility $u(c) = c - \frac{\alpha}{2} c^2$ with $\alpha > 0$ on the range of c where $u' > 0$, or (ii) returns are jointly normally distributed and utility is any increasing concave function.

Corollary 4.5 (The CAPM). *Under Assumption 4.4 the discount factor is affine in the wealth return,*

$$m = a - b R^W, \quad b > 0, \quad (4.4)$$

and expected returns satisfy

$$\mathbb{E}[R^i] = R^f + \beta_{i,W} (\mathbb{E}[R^W] - R^f), \quad \beta_{i,W} = \frac{\text{Cov}(R^i, R^W)}{\text{Var}(R^W)}. \quad (4.5)$$

Proof. Take case (i). The investor's marginal rate of substitution is $m = \delta u'(C_1)/u'(C_0)$, and with quadratic utility $u'(c) = 1 - \alpha c$, so

$$m = \frac{\delta(1 - \alpha C_1)}{1 - \alpha C_0} = \frac{\delta}{1 - \alpha C_0} - \frac{\delta \alpha W_0}{1 - \alpha C_0} R^W =: a - b R^W,$$

using $C_1 = W_0 R^W$. Both a and b are constants known at date 0, and $b > 0$ because $\delta, \alpha, W_0 > 0$ and $u'(C_0) = 1 - \alpha C_0 > 0$. This is (4.4).

Now apply Theorem 4.2 with the single factor $f = R^W$ and coefficient $-b$:

$$\mathbb{E}[R^i] = R^f + \beta_{i,W} \lambda_W, \quad \lambda_W = R^f b \text{Var}(R^W).$$

To identify λ_W without reference to a and b , apply the same relation to the wealth return itself. Since $\beta_{W,W} = 1$,

$$\mathbb{E}[R^W] = R^f + \lambda_W \implies \lambda_W = \mathbb{E}[R^W] - R^f,$$

and substituting back gives (4.5). For case (ii) the same conclusion follows from Stein's lemma: for jointly normal R^i and W_1 , $\text{Cov}(u'(W_1), R^i) = \mathbb{E}[u''(W_1)] \text{Cov}(W_1, R^i)$, so $\text{Cov}(m, R^i)$ is again proportional to $\text{Cov}(R^W, R^i)$ with a negative constant, which is all the argument used. $\square$

Remark 4.6 (What the CAPM actually assumes). Corollary 4.5 makes the cost of the model explicit, and it is high. Case (i) requires quadratic utility, which has increasing absolute risk aversion and a satiation point beyond which more wealth is bad. Case (ii) requires normally distributed returns, which is inconsistent with limited liability and with every option payoff. Both require that consumption equals wealth, so that there is no intertemporal problem and no labour income. The empirical failures of the CAPM discussed in Section 7 should be read against this list rather than as a surprise: the model is a two-date approximation whose assumptions are known to be false, and its usefulness is a question of magnitude, not of principle.

4.3 The consumption CAPM

The natural repair is to stop pretending consumption equals wealth. If a representative investor with time-separable utility chooses consumption optimally, the first-order condition of the intertemporal problem identifies the discount factor directly.

Proposition 4.7 (Consumption-based discount factor). *Let a representative investor maximise $u(C_0) + \delta \mathbb{E}[u(C_1)]$ subject to a budget constraint in which any traded payoff x may be bought at price $p(x)$. Then at an interior optimum*

$$m = \delta \frac{u'(C_1)}{u'(C_0)}, \quad (4.6)$$

and with constant relative risk aversion $u(c) = c^{1-\gamma}/(1-\gamma)$,

$$m = \delta \left(\frac{C_1}{C_0} \right)^{-\gamma} = \delta e^{-\gamma \Delta c}. \quad (4.7)$$

Proof. Let the investor purchase ξ units of the payoff x , financed by reducing date-0 consumption. Lifetime utility as a function of ξ is $V(\xi) = u(C_0 - \xi p(x)) + \delta \mathbb{E}[u(C_1 + \xi x)]$. At an interior optimum $V'(0) = 0$, that is

$$-p(x) u'(C_0) + \delta \mathbb{E}[u'(C_1) x] = 0,$$

where differentiation passes inside the expectation by dominated convergence. Dividing by $u'(C_0) > 0$,

$$p(x) = \mathbb{E} \left[\delta \frac{u'(C_1)}{u'(C_0)} x \right],$$

which is (2.6) with m as in (4.6). Substituting $u'(c) = c^{-\gamma}$ gives (4.7). $\square$

Two features of (4.7) deserve emphasis. First, m is high exactly when consumption growth is low, so Proposition 3.1 says that assets paying badly in recessions must offer higher expected returns. This is the economic content the arbitrage framework of Section 2 could not supply. Second, the model has only two free parameters, δ and γ , which is what makes it falsifiable and, as Section 5 shows, what causes it to fail.

Proposition 4.8 (Lognormal consumption CAPM). *Suppose m is given by (4.7) and that Δc and the log return $r^i = \ln R^i$ are jointly normally distributed. Then*

$$r^f = -\ln \delta + \gamma g - \frac{1}{2} \gamma^2 \sigma_c^2, \quad (4.8)$$

$$\mathbb{E}[r^i] - r^f + \frac{1}{2} \sigma_i^2 = \gamma \text{Cov}(\Delta c, r^i) = \gamma \rho_{ci} \sigma_c \sigma_i, \quad (4.9)$$

where $g = \mathbb{E}[\Delta c]$, $\sigma_c = \sigma(\Delta c)$ and $\sigma_i = \sigma(r^i)$.

Proof. For a jointly normal pair, $\ln \mathbb{E}[e^Z] = \mathbb{E}[Z] + \frac{1}{2} \text{Var}(Z)$. Take logs of $1 = \mathbb{E}[mR^i] = \mathbb{E}[e^{\ln \delta - \gamma \Delta c + r^i}]$ to obtain

$$0 = \ln \delta - \gamma g + \mathbb{E}[r^i] + \frac{1}{2} (\gamma^2 \sigma_c^2 + \sigma_i^2 - 2\gamma \text{Cov}(\Delta c, r^i)).$$

Applying this to the riskless asset, for which $r^i = r^f$ is constant so $\sigma_i = \text{Cov}(\Delta c, r^i) = 0$, gives (4.8). Subtracting the risk-free case from the general one leaves

$$0 = \mathbb{E}[r^i] - r^f + \frac{1}{2} \sigma_i^2 - \gamma \text{Cov}(\Delta c, r^i),$$

which is (4.9). $\square$

Equations (4.8) and (4.9) are the two equations Section 5 confronts with data, and it is worth reading them before the numbers arrive. The premium in (4.9) is linear in γ and in the consumption–return covariance, and that covariance is small because aggregate consumption is smooth. The risk-free rate in (4.8) is linear in γ through the term γg and quadratic through $-\frac{1}{2} \gamma^2 \sigma_c^2$. Raising γ to fix the first equation therefore moves the second, and the two cannot be satisfied at once.

5 Calibration and the Two Puzzles

5.1 The calibration

The moments used below are the standard post-war US annual real magnitudes reported in the consumption-based literature; they are recorded in Table 2 and are taken as given rather than re-estimated here.

Table 2: Calibration inputs. Annual real US magnitudes of the order reported by Mehra and Prescott [MP] and Campbell [Ca]. These are documented inputs, not estimates produced in this note; the conclusions below depend on their order of magnitude and not on their third decimal.

Symbol	Meaning	Value
g	mean log consumption growth	2.00%
σ_c	standard deviation of log consumption growth	1.35%
$\mathbb{E}[R^e]$	mean equity excess return	6.00%
σ_m	standard deviation of the log equity return	16.70%
ρ_{cm}	correlation of consumption growth with the equity return	0.20
r^f	real risk-free rate	1.00%
SR	implied Sharpe ratio $\mathbb{E}[R^e]/\sigma_m$	0.359

5.2 The equity premium puzzle

Apply Theorem 3.3 first, because it requires nothing beyond Table 2. The market offers a Sharpe ratio of 0.359, so any valid discount factor must satisfy $\sigma(m)/\mathbb{E}[m] \geq 0.359$. For the CRRA kernel (4.7) with lognormal consumption growth, a direct calculation gives

$$\frac{\sigma(m)}{\mathbb{E}[m]} = \sqrt{e^{\gamma^2 \sigma_c^2} - 1} \approx \gamma \sigma_c, \quad (5.1)$$

the approximation holding for small $\gamma \sigma_c$. Setting (5.1) equal to the Sharpe ratio and solving exactly,

$$\gamma = \frac{\sqrt{\ln(1 + \text{SR}^2)}}{\sigma_c} = 25.8, \quad (5.2)$$

against $\text{SR}/\sigma_c = 26.6$ from the approximation. Table 3 reports the coefficient of variation of the CRRA kernel as γ varies, with δ chosen at each γ to reproduce the observed risk-free rate.

Table 3: Coefficient of variation of the CRRA discount factor $m = \delta e^{-\gamma \Delta c}$ under the calibration of Table 2, with δ set at each γ so that $\mathbb{E}[m] = 1/(1 + r^f)$. The Hansen–Jagannathan bound of Theorem 3.3 requires the final column to be at least 0.359.

γ	$\mathbb{E}[m]$	$\sigma(m)$	$\sigma(m)/\mathbb{E}[m]$
1.0	0.9901	0.0134	0.0135
5.0	0.9901	0.0669	0.0676
10.0	0.9901	0.1342	0.1355
20.0	0.9901	0.2716	0.2743
25.8	0.9901	0.3556	0.3591
30.0	0.9901	0.4177	0.4219
50.0	0.9901	0.7359	0.7433

Figure 1 shows the same calculation geometrically. The shaded region is the set of $(\mathbb{E}[m], \sigma(m))$ pairs consistent with the observed Sharpe ratio. The CRRA kernel with δ matched to the risk-free rate traces a vertical line, since $\mathbb{E}[m]$ is pinned at $1/(1 + r^f)$, and it enters the admissible region only at $\gamma = 25.8$. The second locus, with δ held fixed at 0.98, shows what happens if the discount factor is not recalibrated: raising γ drives $\mathbb{E}[m]$ away from $1/(1 + r^f)$, which is the risk-free rate puzzle appearing as a horizontal displacement.

The bound is generous, because it is attained only when m is perfectly correlated with the equity return. Imposing the observed correlation $\rho_{cm} = 0.20$ in (4.9) and solving for the risk

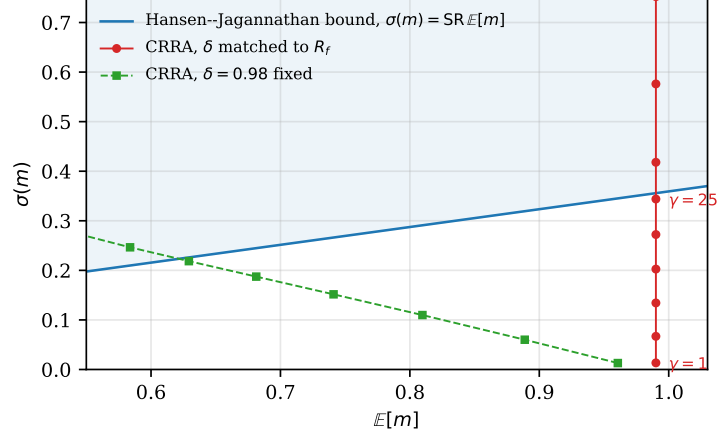


Figure 1: The Hansen–Jagannathan bound of Theorem 3.3 under the calibration of Table 2, with the CRRA discount factor $m = \delta e^{-\gamma \Delta c}$ superimposed for $\gamma = 1, \dots, 50$. Circles: δ chosen at each γ to match the observed risk-free rate, so that $\mathbb{E}[m]$ is fixed. Squares: $\delta = 0.98$ held fixed. The kernel enters the admissible region only at $\gamma = 25.8$. Produced in Python with NumPy and Matplotlib; see Appendix A.

aversion needed to generate a 6% premium,

$$\gamma = \frac{\mathbb{E}[R^e]}{\rho_{cm} \sigma_c \sigma_m} = \frac{0.060}{0.20 \times 0.0135 \times 0.167} = 133.1. \quad (5.3)$$

With $\rho_{cm} = 1$ the same expression gives 26.6. The three figures 25.8, 26.6 and 133.1 are the equity premium puzzle in its three standard forms. For orientation, the values of γ that survive introspection, experimental evidence and the behaviour of insurance markets are of the order of 1 to 10.

5.3 The risk-free rate puzzle

The natural response is to accept a large γ . Equation (4.8) shows why that fails. Writing $\ln \delta = -\varrho$ with ϱ the rate of time preference,

$$r^f = \varrho + \gamma g - \frac{1}{2} \gamma^2 \sigma_c^2. \quad (5.4)$$

The second term is the dominant one at the calibrated magnitudes, because consumption grows at 2% while its volatility is only 1.35%: the precautionary term $-\frac{1}{2} \gamma^2 \sigma_c^2$ is quadratic in γ but multiplied by a very small number. A patient investor with high γ facing growing consumption wants to borrow against the future, and in equilibrium the interest rate must rise enormously to stop him. Table 4 reports the arithmetic.

At the risk aversion needed to explain the equity premium, the model predicts a real risk-free rate between 50% and 107% against an observed 1%. Figure 2 shows both relations. This is Weil’s risk-free rate puzzle [We], and it is the more damaging of the two, because the first puzzle concerns a second moment that might be argued about while the second concerns a first moment that is directly observed.

Remark 5.1 (The puzzles are one puzzle). It is tempting to treat these as two separate failures. They are not. Both follow from the single fact that aggregate consumption is extremely smooth: $\sigma_c = 1.35\%$ against an equity volatility of 16.7%. A smooth kernel cannot be volatile, so by Theorem 3.3 it cannot generate a large Sharpe ratio, and the only way to make it volatile within the CRRA family is to raise γ , which mechanically distorts the mean of the kernel and hence the risk-free rate. Every successful response in the literature attacks that fact directly, by making marginal utility volatile without making consumption volatile. Section 7 surveys the three main routes.

Table 4: The risk-free rate puzzle. Model-implied equity premium from (4.9) and risk-free rate from (5.4), at $\varrho = 2\%$ and the calibration of Table 2. Observed values are a 6.00% premium and a 1.00% risk-free rate. The value $\gamma = 27.8$ is that required to match the premium when $\rho_{cm} = 1$ before the exact lognormal correction; $\gamma = 133.1$ is (5.3).

γ	premium, $\rho_{cm} = 0.20$	premium, $\rho_{cm} = 1$	implied r^f
1.0	0.05%	0.23%	3.99%
2.0	0.09%	0.45%	5.96%
5.0	0.23%	1.13%	11.77%
10.0	0.45%	2.25%	21.09%
27.8	1.25%	6.26%	50.56%
50.0	2.25%	11.27%	79.22%
133.1	6.00%	30.00%	106.77%

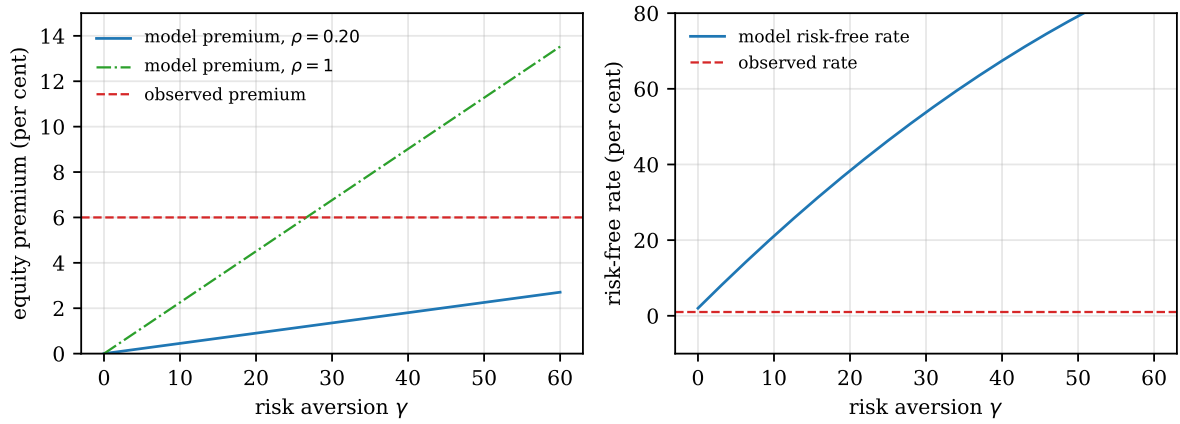


Figure 2: Left: model-implied equity premium from (4.9) against γ , at the observed consumption–return correlation and at $\rho_{cm} = 1$, with the observed premium marked. Right: model-implied real risk-free rate from (5.4) against γ at $\varrho = 2\%$, with the observed rate marked. No single γ reconciles both panels. Produced in Python; see Appendix A.

6 Numerical Implementation

This section does three things. It verifies the pricing equation numerically, so that the implementation of Sections 3 and 4 can be trusted. It then constructs a simulated economy in which the true discount factor has two priced factors, and shows what a single-factor model does when confronted with it. The point of the second exercise is that the failure is entirely predictable from Theorem 4.2: the pricing errors are proportional to the loadings on the omitted factor. All computations are in Python with NumPy, SciPy and Matplotlib; the code is Appendix A, and every random number generator is initialised with the stated seed so that the numbers below reproduce exactly.

6.1 Verifying the pricing equation

Simulate an economy with lognormal consumption growth and a lognormal equity return correlated with it, using the calibration of Table 2. Fix $\gamma = 10$ and $\delta = 0.98$, and set the mean log return by the closed-form condition of Proposition 4.8 so that $\mathbb{E}[mR] = 1$ holds exactly in population. The simulation is then a test of both the algebra and the implementation. With 2 000 000 draws and seed 4242:

$$\mathbb{E}[mR] = 0.999985 \quad (\text{standard error } 0.000138),$$

which is within one standard error of unity. The implied risk-free rate is 23.506% against the closed-form 23.502% of (4.8); the implied arithmetic equity premium is 0.552% against a closed form of 0.558%; and the log premium is 0.451%, matching $\gamma\rho_{cm}\sigma_c\sigma_m = 0.451\%$ from (4.9) to three decimals. Note in passing that the 23.5% risk-free rate at $\gamma = 10$ is Table 4 reappearing inside a simulation: the puzzle is not an artefact of any approximation.

6.2 A simulated cross-section

Construct an economy with two priced factors, f_1 interpreted as the market excess return and f_2 as a second source of priced risk. Monthly parameters are $\mathbb{E}[f_1] = 0.50\%$, $\sigma(f_1) = 4.50\%$, $\mathbb{E}[f_2] = 0.35\%$ and $\sigma(f_2) = 1.80\%$, with the factors independent. Ten test portfolios are formed on an orthogonal 5×2 design: market loadings $b_1 \in \{0.7, 0.9, 1.1, 1.3, 1.5\}$ crossed with second-factor loadings $b_2 \in \{-0.6, +0.6\}$, plus idiosyncratic noise of 1.00% per month. Orthogonality of the design matters, and is the reason for choosing it: if the two loadings were correlated across assets, the cross-section could not separate the factors and the diagnosis below would be ambiguous. The sample is 600 months, seed 20260817.

By construction the true expected excess returns are $\mathbb{E}[R_i^e] = b_{1i}\mathbb{E}[f_1] + b_{2i}\mathbb{E}[f_2]$, so the true CAPM pricing errors are $\alpha_i = b_{2i}\mathbb{E}[f_2] = \pm 0.21\%$ per month, or about 2.5% per year. Table 5 reports what the single-factor time-series regressions recover.

Table 5: Simulated cross-section, $T = 600$ months, seed 20260817. Columns b_1 and b_2 are the true factor loadings; β is the estimated slope on the market factor; α is the estimated intercept of the single-factor time-series regression, with its t -statistic; the final column is the population pricing error $b_2\mathbb{E}[f_2]$. Returns and pricing errors are monthly percentages.

i	b_1	b_2	$\mathbb{E}[R_i^e]$	$\beta_{i,\text{mkt}}$	α_i	$t(\alpha_i)$	true α_i
1	0.70	-0.60	0.084	0.697	-0.230	-3.85	-0.210
2	0.70	0.60	0.605	0.693	0.294	4.87	0.210
3	0.90	-0.60	0.131	0.894	-0.271	-4.44	-0.210
4	0.90	0.60	0.643	0.902	0.237	3.92	0.210
5	1.10	-0.60	0.255	1.086	-0.234	-3.84	-0.210
6	1.10	0.60	0.784	1.088	0.295	4.78	0.210
7	1.30	-0.60	0.308	1.289	-0.272	-4.69	-0.210
8	1.30	0.60	0.818	1.303	0.233	4.09	0.210
9	1.50	-0.60	0.458	1.487	-0.210	-3.53	-0.210
10	1.50	0.60	0.951	1.495	0.279	4.76	0.210

Three things follow from Table 5.

The betas are recovered, the prices are not. Every estimated β is within sampling error of its true loading, so the single-factor model measures exposure to the market correctly. It nevertheless misprices every asset, because exposure to the market is not the only thing that is priced. Estimating betas well and pricing well are different achievements, and only the second is a test of the model.

The errors are systematic, not noise. The sign of α_i tracks the sign of b_{2i} exactly, and the magnitudes cluster around the population value of 0.21%. This is Theorem 4.2 in operation: omitting a priced factor produces pricing errors proportional to loadings on it. In real data this pattern is what a discovered “anomaly” looks like from the inside.

The rejection is statistical as well as economic. The Gibbons–Ross–Shanken statistic [GRS] for the joint hypothesis that all ten intercepts are zero is

$$J = \frac{T - N - K}{N} \left(1 + \frac{\hat{\mu}_f^2}{\hat{\sigma}_f^2} \right)^{-1} \hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha} = 3.61, \quad p = 1.12 \times 10^{-4}, \quad (6.1)$$

on $F(10, 589)$, decisively rejecting the single-factor model.

Re-estimating the same regressions with both factors on the right-hand side removes the problem entirely: the largest absolute intercept falls to 0.066% per month and the largest $|t|$ -statistic to 1.49, so no individual pricing error is significant. The cross-sectional regressions tell the same story. Regressing mean excess returns on estimated CAPM betas gives a slope of 0.458% against a true market premium of 0.500% with $R^2 = 0.201$; the two-factor cross-sectional regression gives slopes of 0.449% and 0.428% against true premia of 0.500% and 0.350%, with $R^2 = 0.991$. Figure 3 plots both.

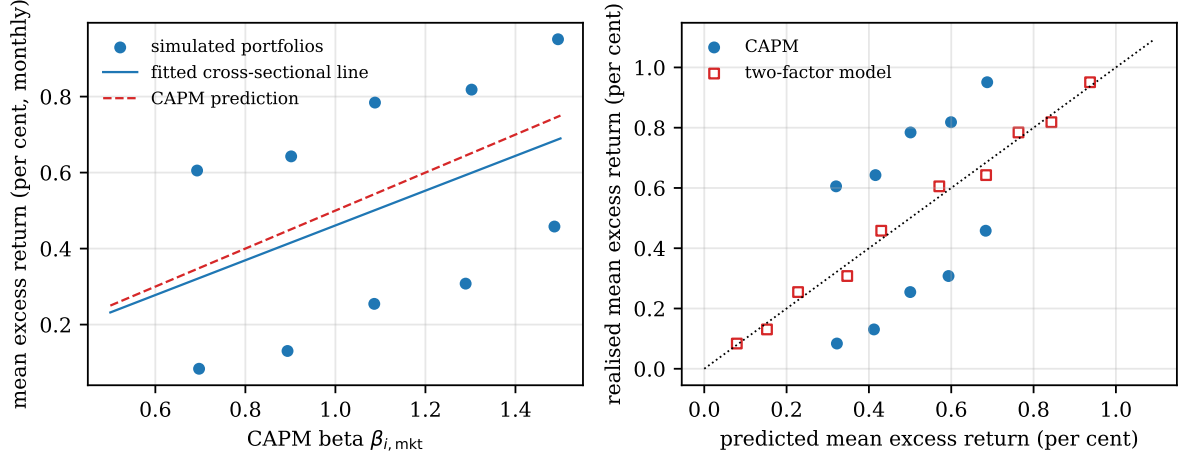


Figure 3: Left: mean excess return against estimated CAPM beta for the ten simulated portfolios, with the fitted cross-sectional line and the CAPM prediction. The vertical spread at each beta is the omitted second factor. Right: realised against predicted mean excess returns under the single-factor and the two-factor model; the dotted line is the 45° line. Monthly percentages, $T = 600$, seed 20260817. Produced in Python; see Appendix A.

Remark 6.1 (What the simulation does and does not establish). The economy above was generated from a two-factor discount factor, so finding that a two-factor model prices it is not evidence for any particular factor structure in real markets. What the exercise establishes is narrower and worth stating precisely: it verifies the estimation machinery on a case where the answer is known, and it demonstrates that the observable signature of an omitted priced factor is a cross-sectionally systematic pattern of alphas aligned with loadings on that factor. In real data one observes the alphas without knowing the true model, which is exactly why a discovered pattern of pricing errors is so hard to interpret: a missing risk factor, a missing conditioning variable and a data-mined coincidence all look identical in Table 5.

7 Where the Model Fails, and the Standard Responses

The framework of Sections 2 to 4 is not in question: it follows from no arbitrage and from a first-order condition. What fails is the specification of m . Three failures are worth separating.

The CAPM fails empirically. Portfolios sorted on size, book-to-market, momentum, profitability and investment display precisely the pattern of Table 5: alphas that are systematic in the sorting variable rather than random. Fama and French [FF] respond by adding factors, which by Theorem 4.2 amounts to adding terms to m . The three- and five-factor models fit far better than the CAPM, but they are descriptive: the factors are constructed from the returns they explain, so the exercise does not by itself identify what makes those factors risky.

The consumption model fails quantitatively. Section 5 recorded the two puzzles, and Remark 5.1 traced both to the smoothness of aggregate consumption. The three main responses each break the link between consumption volatility and marginal-utility volatility in a different

way. Campbell and Cochrane [CC] make risk aversion time-varying through an external habit, so that marginal utility is volatile near the habit even when consumption is smooth. Bansal and Yaron [BY] introduce a small persistent component in consumption growth, so that a modest shock today revises lifetime expectations substantially. Epstein and Zin [EZ] break the mechanical tie between risk aversion and the elasticity of intertemporal substitution that CRRA utility imposes, which is what allows the risk-free rate in (5.4) to be controlled independently of the premium in (4.9). Each of these succeeds at the calibration it was designed for; none commands consensus.

Conditioning information is not innocuous. Everything above was written with unconditional moments. The pricing equation (2.6) in fact holds conditional on date-0 information, and conditional and unconditional beta representations are not equivalent when betas or premia vary over time. A model can hold conditionally and fail unconditionally, and the reverse. Empirical rejections of the CAPM are therefore not unambiguously rejections of the theory rather than of the assumption that betas are constant.

Remark 7.1 (What survives). None of this undermines the framework. Theorem 2.6 holds regardless of which model of marginal utility is correct, and Theorem 3.3 is a valid diagnostic against any candidate. The useful position is that $p = \mathbb{E}[mx]$ is not a theory to be tested but a language in which theories are written, and that the open question is the specification of m , not the structure of pricing.

8 Conclusion

Three conclusions follow from the argument above.

First, the existence of a strictly positive discount factor is equivalent to the absence of arbitrage, and nothing else is required to obtain it. Uniqueness is a separate property, equivalent to completeness. Every result in Section 3 — the risk premium, the beta representation, the Hansen–Jagannathan bound, the mean–variance frontier — follows from that one equation together with the definition of covariance.

Second, the CAPM is not an alternative to this framework but an instance of it, obtained by asserting that the discount factor is affine in the return on aggregate wealth. Since Theorem 4.2 shows that any linear factor model is equivalent to a beta representation, the empirical success of a beta relation is not evidence for a theory. The economic content lies in the prior specification of the factor, and the CAPM’s is purchased with assumptions — quadratic utility or normal returns, and consumption equal to wealth — that are known to be false.

Third, when the discount factor is given genuine economic content as a marginal rate of substitution, the model fails quantitatively and in a way that is easy to state: matching the equity premium requires $\gamma = 25.8$ at the volatility bound and 133.1 at the observed consumption–return correlation, and either figure implies a risk-free rate of tens of per cent against an observed one per cent. Both failures are consequences of a single fact, the smoothness of aggregate consumption, and every serious response in the literature is an attempt to make marginal utility volatile without making consumption volatile.

The numerical work of Section 6 adds one further point. In a simulated economy where a priced factor is omitted, the resulting pricing errors are systematic in the loadings on the omitted factor and are rejected decisively by a joint test, while the betas themselves are estimated accurately throughout. Estimating exposures well is not the same as pricing well, and only the second is a test of a model.

A Python Implementation

The listing below produces every number, table entry and figure in Sections 5 and 6. Seeds are stated in the code; running it as given reproduces the reported values exactly.

```

1  """
2  Numerical work for: The Stochastic Discount Factor and the CAPM.
3  Every figure and every table entry quoted in the paper is produced here.
4  """
5  import numpy as np
6  import matplotlib
7  matplotlib.use("Agg")
8  import matplotlib.pyplot as plt
9
10 plt.rcParams.update({"font.family": "serif", "font.size": 9,
11                     "axes.grid": True, "grid.alpha": 0.3, "figure.dpi": 200})
12
13 # ----- calibration
14 g = 0.0200 # mean log consumption growth
15 sig_c = 0.0135 # s.d. of log consumption growth
16 mu_e = 0.0600 # mean log equity excess return
17 sig_m = 0.1670 # s.d. of log equity return
18 rho_cm = 0.2000 # corr(consumption growth, equity return)
19 rf_obs = 0.0100 # observed real risk-free rate
20 SR = mu_e / sig_m
21 print("=== Table 5.1 calibration ===")
22 print(f"g={g} sig_c={sig_c} mu_e={mu_e} sig_m={sig_m} rho={rho_cm} rf={rf_obs}")
23 print(f"Sharpe ratio = {SR:.4f}")
24
25 # ----- Section 5: implied gamma
26 gam_hj_perfect = SR / sig_c # HJ bound, |rho|=1
27 gam_lognormal = mu_e / (rho_cm * sig_c * sig_m) # lognormal CCAPM, rho as observed
28 gam_lognormal_1 = mu_e / (1.0 * sig_c * sig_m)
29 print("\n=== Table 5.2 implied risk aversion ===")
30 print(f"HJ bound (|rho|=1) gamma >= {gam_hj_perfect:.1f}")
31 print(f"lognormal CCAPM, rho=1 gamma = {gam_lognormal_1:.1f}")
32 print(f"lognormal CCAPM, rho=0.20 gamma = {gam_lognormal:.1f}")
33
34 def rf_model(gamma, delta=0.02):
35     return delta + gamma * g - 0.5 * gamma ** 2 * sig_c ** 2
36
37 def prem_model(gamma, rho=rho_cm):
38     return gamma * rho * sig_c * sig_m
39
40 print("\n=== Table 5.3 risk-free rate puzzle (delta = 0.02) ===")
41 for gm in [1, 2, 5, 10, 27.8, 50, 133.1]:
42     print(f"gamma={gm:6.1f} premium(rho=0.2)={100*pre_model(gm):6.2f}% "
43           f"premium(rho=1)={100*pre_model(gm,1.0):7.2f}% rf={100*rf_model(gm):8.2f}%")
44
45 # ----- Figure 5.1: Hansen-Jagannathan
46 # Excess-return bound: sigma(m)/E[m] >= |E[Re]|/sigma(Re) = Sharpe ratio.
47 Em_grid = np.linspace(0.55, 1.03, 400)
48 bound = SR * Em_grid
49
50 def crra_moments(gamma, delta):
51     Em = delta * np.exp(-gamma * g + 0.5 * gamma ** 2 * sig_c ** 2)
52     Em2 = delta ** 2 * np.exp(-2 * gamma * g + 2 * gamma ** 2 * sig_c ** 2)
53     return Em, np.sqrt(Em2 - Em ** 2)
54
55 def delta_matching_rf(gamma):
56     """Time discount factor that reproduces the observed risk-free rate."""
57     return (1.0 / (1 + rf_obs)) / np.exp(-gamma * g + 0.5 * gamma ** 2 * sig_c ** 2)
58
59
60 gam_pts = np.array([1, 5, 10, 15, 20, 25, 30, 40, 50], dtype=float)
61 pts_cal = np.array([crra_moments(gm, delta_matching_rf(gm)) for gm in gam_pts])
62 pts_fix = np.array([crra_moments(gm, 0.98) for gm in gam_pts])
63
64 gam_star = np.sqrt(np.log(1 + SR ** 2)) / sig_c
65 print(f"ngamma satisfying the HJ bound exactly: {gam_star:.2f}")
66 print(f"check: E[m]={pts_cal[0,0]:.4f} (target {1/(1+rf_obs):.4f})")
67
68 fig, ax = plt.subplots(figsize=(4.9, 3.1))
69 ax.plot(Em_grid, bound, color="tab:blue", lw=1.2,
70         label=r"Hansen--Jagannathan bound, $\sigma(m)=\mathrm{SR}\backslash,\mathrm{E}[m]$" )
71 ax.fill_between(Em_grid, bound, 0.9, color="tab:blue", alpha=0.08)
72 ax.plot(pts_cal[:, 0], pts_cal[:, 1], "o-", color="tab:red", ms=3.2, lw=0.9,
73         label=r"CRRA, $\delta$ matched to $R_f$")

```

```

74 ax.plot(pts_fix[:, 0], pts_fix[:, 1], "s--", color="tab:green", ms=3.0, lw=0.9,
75         label=r"CRRA,  $\delta=0.98$  fixed")
76 for gm, (x, y) in zip(gam_pts, pts_cal):
77     if gm in (1, 25, 50):
78         ax.annotate(rf"$\gamma={int(gm)}$", (x, y), textcoords="offset points",
79                   xytext=(6, -3), fontsize=7.5, color="tab:red")
80 for gm, (x, y) in zip(gam_pts, pts_fix):
81     if gm in (50,):
82         ax.annotate(rf"$\gamma={int(gm)}$", (x, y), textcoords="offset points",
83                   xytext=(4, 2), fontsize=7.5, color="tab:green")
84 ax.set_xlim(0.55, 1.03); ax.set_ylim(0, 0.75)
85 ax.set_xlabel(r"$\mathbb{E}[m]$"); ax.set_ylabel(r"$\sigma(m)$")
86 ax.legend(frameon=False, fontsize=7.5, loc="upper left")
87 fig.tight_layout(); fig.savefig("fig_hj.pdf")
88
89 print("\n=== Table 5.2b CRRA volatility of the SDF ===")
90 for gm in [1, 5, 10, 20, 25.8, 30, 50]:
91     Em, sm = crra_moments(gm, delta_matching_rf(gm))
92     print(f"gamma={gm:6.1f} E[m]={Em:.4f} sigma(m)={sm:.4f} ratio={sm/Em:.4f} "
93           f"bound {SR:.4f}")
94
95 # ----- Figure 5.2: premium and risk-free puzzle
96 gg = np.linspace(0, 60, 400)
97 fig, ax = plt.subplots(1, 2, figsize=(7.4, 2.8))
98 ax[0].plot(gg, 100 * prem_model(gg, rho_cm), color="tab:blue", lw=1.2,
99           label=r"model premium,  $\rho=0.20$ ")
100 ax[0].plot(gg, 100 * prem_model(gg, 1.0), color="tab:green", lw=1.0, ls="-. ",
101           label=r"model premium,  $\rho=1$ ")
102 ax[0].axhline(100 * mu_e, color="tab:red", ls="--", lw=1.0, label="observed premium")
103 ax[0].set_xlabel(r"risk aversion  $\gamma$ "); ax[0].set_ylabel("equity premium (per cent)")
104 ax[0].set_ylim(0, 15); ax[0].legend(frameon=False, fontsize=7.5)
105 ax[1].plot(gg, 100 * rf_model(gg), color="tab:blue", lw=1.2, label="model risk-free rate")
106 ax[1].axhline(100 * rf_obs, color="tab:red", ls="--", lw=1.0, label="observed rate")
107 ax[1].set_xlabel(r"risk aversion  $\gamma$ "); ax[1].set_ylabel("risk-free rate (per cent)")
108 ax[1].set_ylim(-10, 80); ax[1].legend(frameon=False, fontsize=7.5)
109 fig.tight_layout(); fig.savefig("fig_puzzle.pdf")
110
111 # ----- Section 6: simulated cross-section
112 rng = np.random.default_rng(20260817)
113 T_obs = 600 # monthly observations, 50 years
114 n_assets = 10
115 lam1, lam2 = 0.0050, 0.0035 # monthly factor risk premia
116 s1, s2 = 0.0450, 0.0180 # factor volatilities
117 # orthogonal 5 x 2 design: five market loadings crossed with two second-factor loadings
118 b1 = np.repeat(np.array([0.70, 0.90, 1.10, 1.30, 1.50]), 2)
119 b2 = np.tile(np.array([-0.60, 0.60]), 5)
120 s_eps = 0.010
121
122 f1 = lam1 + s1 * rng.standard_normal(T_obs)
123 f2 = lam2 + s2 * rng.standard_normal(T_obs)
124 eps = s_eps * rng.standard_normal((T_obs, n_assets))
125 Re = b1 * (f1[:, None] - lam1) + b2 * (f2[:, None] - lam2) + b1 * lam1 + b2 * lam2 + eps
126
127 # true (population) values
128 true_mean = b1 * lam1 + b2 * lam2
129 true_beta_capm = (b1 * s1 ** 2) / (s1 ** 2) # = b1 since factors independent
130 true_alpha = true_mean - true_beta_capm * lam1
131
132 # time-series CAPM regressions on the market factor only
133 X = np.column_stack([np.ones(T_obs), f1])
134 XtXi = np.linalg.inv(X.T @ X)
135 coef = XtXi @ X.T @ Re
136 resid = Re - X @ coef
137 s2res = (resid ** 2).sum(axis=0) / (T_obs - 2)
138 se = np.sqrt(np.outer(np.diag(XtXi), s2res))
139 alpha, beta = coef[0], coef[1]
140 t_alpha = alpha / se[0]
141
142 print("\n=== Table 6.1 simulated cross-section (seed 20260817, T=600) ===")
143 print(" i b1 b2 E[Re] beta alpha t(alpha) true alpha")
144 for i in range(n_assets):
145     print(f"{i+1:2d} {b1[i]:5.2f} {b2[i]:6.2f} {100*Re[:, i].mean():7.3f} {beta[i]:6.3f} "
146           f"{100*alpha[i]:7.3f} {t_alpha[i]:8.2f} {100*true_alpha[i]:11.3f}")

```

```

147
148 # GRS-style joint statistic
149 Sig_res = np.cov(resid.T, bias=True)
150 Sig_res += 1e-12 * np.eye(n_assets)
151 mu_f, var_f = f1.mean(), f1.var()
152 grs = (T_obs - n_assets - 1) / n_assets * (1 + mu_f ** 2 / var_f) ** -1 * \
153     (alpha @ np.linalg.solve(Sig_res, alpha))
154 from scipy.stats import f as fdist
155 p_grs = 1 - fdist.cdf(grs, n_assets, T_obs - n_assets - 1)
156 print(f"GRS statistic = {grs:.2f}, p-value = {p_grs:.3e}")
157
158 # two-factor time-series regression: alphas should vanish
159 X2 = np.column_stack([np.ones(T_obs), f1, f2])
160 coef2 = np.linalg.inv(X2.T @ X2) @ X2.T @ Re
161 resid2 = Re - X2 @ coef2
162 s2res2 = (resid2 ** 2).sum(axis=0) / (T_obs - 3)
163 se2 = np.sqrt(np.outer(np.diag(np.linalg.inv(X2.T @ X2)), s2res2))
164 t_alpha2 = coef2[0] / se2[0]
165 print(f"two-factor model: max |alpha| = {100*np.abs(coef2[0]).max():.4f}%, "
166       f"max |t| = {np.abs(t_alpha2).max():.2f}")
167
168 # cross-sectional regression of mean excess returns on CAPM betas
169 A = np.column_stack([np.ones(n_assets), beta])
170 gam_cs = np.linalg.lstsq(A, Re.mean(axis=0), rcond=None)[0]
171 r2 = 1 - ((Re.mean(axis=0) - A @ gam_cs) ** 2).sum() / \
172     ((Re.mean(axis=0) - Re.mean()) ** 2).sum()
173 print(f"cross-sectional CAPM: intercept={100*gam_cs[0]:.4f}%, slope={100*gam_cs[1]:.4f}% "
174       f"(true premium {100*lam1:.4f}%), R2={r2:.3f}")
175
176 A2 = np.column_stack([np.ones(n_assets), coef2[1], coef2[2]])
177 gam_cs2 = np.linalg.lstsq(A2, Re.mean(axis=0), rcond=None)[0]
178 r2b = 1 - ((Re.mean(axis=0) - A2 @ gam_cs2) ** 2).sum() / \
179     ((Re.mean(axis=0) - Re.mean()) ** 2).sum()
180 print(f"cross-sectional 2-factor: slopes = {100*gam_cs2[1]:.4f}%, {100*gam_cs2[2]:.4f}% "
181       f"(true {100*lam1:.4f}%, {100*lam2:.4f}%), R2={r2b:.3f}")
182
183 # ----- Figure 6.1: security market line
184 fig, ax = plt.subplots(1, 2, figsize=(7.4, 2.9))
185 bb = np.linspace(0.5, 1.5, 10)
186 ax[0].scatter(beta, 100 * Re.mean(axis=0), s=18, color="tab:blue", zorder=3,
187              label="simulated portfolios")
188 ax[0].plot(bb, 100 * gam_cs[0] + 100 * gam_cs[1] * bb, color="tab:blue", lw=1.0,
189            label="fitted cross-sectional line")
190 ax[0].plot(bb, 100 * lam1 * bb, ls="--", color="tab:red", lw=1.0,
191            label="CAPM prediction")
192 ax[0].set_xlabel(r"CAPM beta  $\beta_i, \mathrm{mkt}$ ")
193 ax[0].set_ylabel("mean excess return (per cent, monthly)")
194 ax[0].legend(frameon=False, fontsize=7.5)
195 ax[1].scatter(100 * (A @ gam_cs), 100 * Re.mean(axis=0), s=18, color="tab:blue",
196              label="CAPM")
197 ax[1].scatter(100 * (A2 @ gam_cs2), 100 * Re.mean(axis=0), s=18, marker="s",
198              facecolors="none", edgecolors="tab:red", label="two-factor model")
199 lims = [0, 100 * Re.mean(axis=0).max() * 1.15]
200 ax[1].plot(lims, lims, color="k", lw=0.8, ls=":")
201 ax[1].set_xlabel("predicted mean excess return (per cent)")
202 ax[1].set_ylabel("realised mean excess return (per cent)")
203 ax[1].legend(frameon=False, fontsize=7.5)
204 fig.tight_layout(); fig.savefig("fig_sml.pdf")
205
206 # ----- Section 6: SDF Monte Carlo check
207 rng2 = np.random.default_rng(4242)
208 n_sim = 2_000_000
209 gamma_true, delta_true = 10.0, 0.98
210 Zc = rng2.standard_normal(n_sim)
211 Zi = rng2.standard_normal(n_sim)
212 dc = g + sig_c * Zc
213 r_m = rho_cm * sig_m * Zc + np.sqrt(1 - rho_cm ** 2) * sig_m * Zi
214 # choose the mean log return so that  $E[mR] = 1$  exactly, then verify by simulation
215 mu_r = -np.log(delta_true) + gamma_true * g - 0.5 * gamma_true ** 2 * sig_c ** 2 \
216     - 0.5 * sig_m ** 2 + gamma_true * rho_cm * sig_c * sig_m
217 R = np.exp(mu_r + r_m)
218 m = delta_true * np.exp(-gamma_true * dc)
219 pv = (m * R).mean()

```

```

220 se_pv = (m * R).std() / np.sqrt(n_sim)
221 print("\n=== Section 6.4 Monte Carlo check of E[mR]=1 (seed 4242, n=2,000,000) ===")
222 print(f"gamma={gamma_true}, delta={delta_true}: E[mR] = {pv:.6f} (s.e. {se_pv:.6f})")
223 print(f"implied risk-free rate = {100*(1/m.mean()-1):.3f}% "
224       f"(closed form {100*(np.exp(rf_model(gamma_true, -np.log(delta_true)))-1):.3f}%)")
225 Rf_cf = 1 / (delta_true * np.exp(-gamma_true * g + 0.5 * gamma_true ** 2 * sig_c ** 2))
226 prem_cf = np.exp(mu_r + 0.5 * sig_m ** 2) - Rf_cf
227 print(f"implied equity premium = {100*(R.mean()-1/m.mean()):.3f}% "
228       f"(closed form {100*prem_cf:.3f}%)")
229 print(f"log premium E[r]-r_f+0.5 var = {100*(mu_r - np.log(Rf_cf) + 0.5*sig_m**2):.3f}% "
230       f"(gamma*rho*sig_c*sig_m = {100*gamma_true*rho_cm*sig_c*sig_m:.3f}%)")
231 print("done")

```

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